Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/05/2024
PI Name:	Weber, Richard Ph.D.		
Project Title:	Microgravity Investigation of Thermophysical Properties of Supercooled Molten Metal Oxides		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	847-612-8597
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City:	Arlington Heights	**State:**	IL
Zip Code:	60004-7708	**Congressional District:**	8
Comments:	NOTE: Formerly with Containerless Research, Inc., until 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	05/21/2017	**End Date:**	05/20/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 05/20/2025 per NSSC information (Ed., 6/21/23) NOTE: End date changed to 05/20/2024 per NSSC information (Ed., 6/21/23) NOTE: End date changed to 05/20/2023 per NSSC information (Ed., 8/25/22) NOTE: New grant number 80NSSC18K0059 awarded with period of performance of 11/9/2017-5/20/2022; added to original grant since is continuation of the work (Ed., 5/18/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K0059 ; NNX17AH73G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project is investigating the thermophysical properties of molten metal oxides. These materials are of critical importance in a wide range of applications including functional glasses, batteries, next generation propulsion systems, lasers and optical products. An increased understanding of oxide materials is of fundamental value in materials science and applied value in developing means to harness planetary materials for use in space exploration. Fundamental research provides an essential underpinning for transformational discovery and development of major new applications. This project aims to measure the density, thermal expansion coefficient, viscosity, and surface tension of a variety of molten oxides, some of which can form glass. Containerless conditions are essential in order to achieve deep supercooling without extrinsic heterogeneous nucleation of crystals. This requirement will be met by using the Japanese Aerospace Exploration Agency (JAXA) Electrostatic Levitation Furnace (ELF). The ELF instrument is an electrostatic levitator designed for work on metal oxides at temperatures up to about 2500 ºC in containerless conditions. By working in microgravity, nearly quiescent molten samples can be obtained. Induced oscillations of levitated liquid droplets is used to study their fluid properties. Image analysis reveals the oscillation amplitude and power spectrum, from which the required properties (surface tension and viscosity) are calculated. The materials of primary interest are molten metal oxides, many of which form fragile liquids that can be made into glass provided the melts can be deeply supercooled. In some cases there is already considerable data on the atomic structure of these melts. An important goal of this research is to correlate the structure and properties (in particular viscosity and density) to advance models of the processing of materials. These models are important in advancing basic science of liquids and glasses and in development of high value-added applied materials. Samples returned from the flight experiments are being investigated by a variety of techniques and in collaboration with several research groups at US universities and National Laboratories. ED. NOTE 5/18/2020: New grant number 80NSSC18K0059 awarded with period of performance of 11/9/2017-5/20/2024; added to original grant since is continuation of the work.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is investigating molten metal oxides. These materials are of critical importance in a wide range of applications including functional glasses, batteries, next generation propulsion systems, lasers and optical products. An increased understanding of oxide materials is of fundamental value in materials science and applied value in developing means to harness planetary materials for use in space exploration. Fundamental research provides an essential underpinning for transformational discovery and development of major new applications. The project work includes outreach to undergraduate students. The experiments in microgravity are measuring essential thermophysical property data that can be used to advance understanding of how to process and optimize materials performance both on Earth and for space exploration.
	This project is investigating the thermophysical properties of molten metal oxides. These materials are of critical importance in a wide range of applications, including functional glasses, batteries, next-generation propulsion systems, lasers, and optical products. An increased understanding of oxide materials is of fundamental value in materials science and applied value to technologies for processing planetary materials in space exploration. Fundamental research provides an essential underpinning for transformational discovery and the development of major new applications. This project is measuring the density, thermal expansion coefficient, viscosity, and surface tension of a variety of molten oxides, some of which can form glass. Containerless conditions are essential to achieve deep supercooling without extrinsic heterogeneous nucleation of crystals. This requirement is met by using the Japanese Aerospace Exploration Agency (JAXA) Electrostatic Levitation Furnace (ELF). The ELF instrument is an electrostatic levitator designed for work on metal oxides at temperatures up to about 2500 ºC in containerless conditions. By working in microgravity, nearly quiescent molten samples can be obtained. Induced oscillations of levitated liquid droplets are used to study their fluid properties. Image analysis reveals the oscillation amplitude and power spectrum, from which the required properties (surface tension and viscosity) are calculated. The materials of primary interest are molten metal oxides, many of which form fragile liquids that can be made into glass, provided the melts can be deeply supercooled. In some cases, there is already considerable data on the atomic structure of these melts. An important goal of this research is to correlate the structure and properties (in particular, viscosity and density) to advance models for the processing of materials. These models are important in advancing basic science of liquids and glasses and in the development of high value-added applied materials. Samples returned from the flight experiments are being investigated by a variety of techniques and in collaboration with several research groups at U.S. universities and national laboratories. 1. Status: The project team includes Materials Development Inc. (MDI) scientists; and Dr. Shinji Kohara, National Institute for Materials Science, Japan; and Dr. Takehiko Ishikawa, JAXA, Japan. Experiments on ELF are performed by Dr. Chihiro Koyama and coordinated by Dr. Rina Shimonishi from JAXA. The collaborators from Japan are not funded by this grant. The grant was awarded on 5/21/2017 and assigned a new contract number in 11/2017. First-year funding was at a level sufficient to scope the work and perform preliminary ground-based experiments in preparation for a more detailed experimental campaign during the subsequent years. Second-year funding was at a level sufficient for planning and preparation for the Science Concept Review (SCR). The SCR was successfully completed in March 2020, and the remaining issues from the SCR were addressed soon afterwards. The first round of flight experiments was started in 11/2021 and completed in 3/2022. This work included experiments on more than 30 samples. Due to some operational issues, some samples could not be run. In an effort to complete the investigation, JAXA provided a fourth sample holder, and MDI provided additional samples. The fourth sample holder was sent to the International Space Station (ISS) on SpaceX CRS-27 (SpX-27) in 3/2023, and experiments were performed in 2&3/2024. MDI was granted an extension of the period of performance due to delays in the implementation of ISS experiments. Starting in 8/2022, the NASA flight support staff and MDI investigators meet once a month via a Teams meeting hosted by Dr. Ellen Rabenberg at NASA Marshall Space Flight Center. The meeting maintains communications and provides for discussion of technical or project management matters that need to be addressed. 2. Technical Activities: Activities during the seventh year included: (i) running experiments on samples that were delivered in 2023, (ii) analyzing data and recovered samples, (iii) making ground-based structure and property measurements on glass and liquid samples, and (iv) publishing results. (i) Flight experiments were run in the JAXA ELF during the period 11/2021-3/2022 and 2&3/2024. Initially, five

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samples each of nine different compositions were launched. An additional holder of samples was launched in 3/2023. These samples were run in ELF in early 2024.

A precision stainless steel calibration sphere (identified as SUS) was measured at the start and end of each sample holder set. The methodology for calculating sample volume compared to the calibration sphere was refined, based on a collaboration with Robert Hyers and Richard Bradshaw. Following these modifications to the density analysis algorithm, all sample density analyses were re-run using mass data from recovered samples and internal porosity corrections obtained from post-flight X-ray tomography. Viscosity and surface tension data were obtained from the drop oscillation spectra. These data were reduced using programs provided by JAXA with filter settings selected based on analysis of the error sources and effects of filter bandpass and frequency range [1]. Once the data processing approach was developed, it was applied consistently to all data sets in order to avoid introducing subjective bias.

During the 2/2024 experiments, Richard Weber and Stephen Wilke traveled to the JAXA facility in Tsukuba to participate in the work. This single visit to JAXA was important to develop a full understanding of the experimental work. The fourth holder was very successful; 16 of 19 samples were levitated and melted in ELF. All compositions heated as intended.

We are currently waiting for return of the samples from the last campaign. Based on the initial sample mass, a preliminary measurement of the density of calcium silicate has been made.

(ii) Analyzing data is an ongoing process as new data and details are available. Preliminary analyses of density from the early 2024 experiments have been performed for all compositions. These will be corrected when we receive the flight samples, weigh them, and, if necessary, perform tomography to enable correction for any internal voids. Models are being used to help with data analysis and to help understand results. This work includes a collaboration with Professor Collin Wilkinson at Alfred University. He is an expert in modeling melt properties from first principles. Comparing modeled data with measured structure factors will provide a check on the quality of the model fit, lending additional weight to the value of the data. The model will then be used for calculating viscosity, to compare with the ELF flight measurements. Preliminary results indicate that with suitable pair potentials, the model reproduces the measured viscosity of samples that were measured in the ELF experiments.

(iii) Ground-based tests are in progress to characterize the compositions for flight and to make measurements on the samples recovered from the flight experiments. Flight samples were examined by optical microscopy to determine the overall appearance of the beads. High energy white beam X-ray tomography was performed at the Advanced Photon Source beamline 7 BM-B to determine the interior structure of the samples with a resolution of ca. 5 μm. These measurements are useful to evaluate the presence of bubbles or interior features that might affect the calculated volume (and hence density) of glass samples. In all cases, both flight and ground-based samples were examined to enable comparison of the materials processed in microgravity and on Earth.

For prior sets of samples, non-destructive tests using x-ray and neutron scattering, x-ray tomography, and nuclear magnetic resonance (NMR) were performed. The results of the measurements for molten Yb3Al5O12, one of the Superglass compositions, were also used to refine a structural model using empirical potential structure refinement (EPSR). The structural model is in good agreement with the scattering data and will be published soon.

After completion of non-destructive tests, glassy samples were sent to the Materials of the Universe Laboratory at Arizona State University. There, the enthalpy of the solution was measured to obtain the heat of vitrification. Preliminary analysis indicates that the glass thermal properties are similar to those made on Earth.

Work is in progress at NASA Marshall Space Flight Center to add capabilities to the ground-based electrostatic levitation laboratory (ESL) to enable measurements of density and synthesis of glasses. This will provide additional glass samples that can be compared to those made in microgravity.

Ground based experiments at MDI have included pulling fibers from selected compositions to investigate the viscosity in a range that cannot be accessed for supercooled liquids by containerless drop oscillation or contained submerged bob techniques.

(iv) Presenting and publishing results is an important aspect of the research. Details of publications are given below.

Publications/presentations: Several papers have been published based on the ISS experiments. Talks were presented at the ASGSR meeting in Washington, DC (Weber), American Ceramic Society Glass and Optical Materials meeting in Las Vegas, May 2024 (Weber invited, Wilke), 22nd Symposium on Thermophysical Properties in Boulder, CO, June 2024 (Weber) and Non-Crystalline Materials (NCM-15), Cambridge, UK, July 2024 (Weber, Wilke). An abstract is pending for the 2024 ASGSR meeting.

Peer-reviewed scientific papers were published. [Ed. Note: See Cumulative Bibliography for complete citations.]

R. Weber, S.K. Wilke, and C.J. Benmore, “Containerless Techniques for in-situ X-Ray Measurements on Materials in Extreme Conditions,” J. Phys. Soc. Jpn., 91, 091008 (2022). S.K. Wilke, C.J. Benmore, J. Ilavsky, R.E. Youngman, A. Rezikyan, M.P. Carson, V. Menon and R. Weber, “Phase separation in mullite-composition glass,” Sci. Rep., 12, 17687 (2022).

S.K. Wilke, C.J. Benmore, J. Ilavsky, R.E. Youngman, A. Rezikyan, M.P. Carson, V. Menon and R. Weber, “Revisiting metastable immiscibility in SiO2–Al2O3: Structure and phase separation of supercooled liquids and glasses,” J. Am. Ceram. Soc., 106, 2820 (2023). Editor’s Choice paper.

Papers on the flight data are in development.

3. Plans: Plans for the next project period are: (i) continue data analysis, (ii) continue sample analysis and characterization, (iii) publish/present results.

JAXA has offered to provide additional ELF experiments to complete the set of measurements as proposed. The ability to do this is contingent on funding of MDI’s portion of the work. MDI’s current funding is running short. As a result, it will be difficult to complete analysis and publication of existing data.

4. References cited:

[1] S.K. Wilke, A. Al-Rubkhi, V. Menon, J. Rafferty, C. Koyama, T. Ishikawa, H. Oda, R.W. Hyers, R.C. Bradshaw,

	A.L. Kastengren, S. Kohara, M. SanSoucie, B.L. Phillips and R. Weber, “Measuring the density, viscosity, and surface tension of molten titanates using electrostatic levitation in microgravity,” Appl. Phys. Lett., 124, 264102 (2024).
Bibliography Type:	Description: (Last Updated: 06/06/2026)
Articles in Peer-reviewed Journals	Wilke SK, Al-Rubkhi A, Menon V, Rafferty J, Koyama C, Ishikawa T, Oda H, Hyers RW, Bradshaw RC, Kastengren AL, Kohara S, SanSoucie M, Phillips B, Weber R. "Measuring the density, viscosity, and surface tension of molten titanates using electrostatic levitation in microgravity." Appl. Phys. Lett. 2024 Jun 24;124(26). https://doi.org/10.1063/5.0198322 , Jun-2024
Articles in Peer-reviewed Journals	Benmore CJ, Bogle R, Wilke SK, Weber R, Neuefeind J, Costa G. "The structure of CaO–MgO–Al2O3–SiO2 melts and glasses doped with FeOX–NiO." Journal of the American Ceramic Society. 2024 May 5;107(9):6323-33. https://doi.org/10.1111/jace.19856 , May-2024